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Brain responses during auditory word recognition vary with reading ability in Chinese school-age children

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Abstract

While the close relationship between the brain system for speech processing and reading development is well-documented in alphabetic languages, whether and how such a link exists in children in a language without systematic grapheme-phoneme correspondence has not been directly investigated. In the present study, we measured Chinese children's brain activation during an auditory lexical decision task with functional magnetic resonance imaging. The results showed that brain areas distributed across the temporal and frontal lobes activated during spoken word recognition. In addition, the left occipitotemporal cortex (OTC) was recruited, especially under the real word condition, thus confirming the involvement of this orthographic-related area in spoken language processing in Chinese children. Importantly, activation of the left temporoparietal cortex (TPC) in response to words and pseudowords was positively correlated with children's reading ability, thus supporting the salient role phonological processing plays in Chinese reading in the developing brain. Furthermore, children with higher reading scores also increasingly recruited the left anterior OTC to make decisions on the lexical status of pseudowords, indicating that higher-skill children tend to search abstract lexical representations more deeply than lower-skill children in deciding whether spoken syllables are real. In contrast, the precuneus was more related to trial-by-trial reaction time in lower-skill children, suggesting that effort-related neural systems differ among pupils with varying reading abilities. Taken together, these findings suggest a strong link between the neural correlates of speech processing and reading ability in Chinese children, thus supporting a universal basis underlying reading development across languages.

KEYWORDS

auditory word recognition, functional magnetic resonance imaging, individual differences, occipitotemporal cortex, reading ability, temporoparietal cortex

1 | INTRODUCTION

Children develop fluent reading skills by successfully combining brain circuits for processing spoken language and specific visual input (orthography). This concept has been proposed and well-documented in alphabetic languages (Blomert, 2011; Dehaene et al., 2010). Recently, a cross-cultural study demonstrated that the convergence

of brain networks for speech and orthographic processing is a neural hallmark of proficient reading shared by healthy adult readers in four contrasting writing systems, namely, Spanish, English, Hebrew, and Chinese, suggesting recruiting the speech brain network for reading could be a necessarily language-universal step in typical reading acquisition (Rueckl et al., 2015). Therefore, knowledge about the relationship between speech processing and reading ability, and the underlying



neural substrates in the developing brain, can provide valuable insights into reading acquisition and improve reading instruction across cultures.

To date, studies have suggested that a reciprocal relationship occurs between speech and reading at the brain level (Dehaene et al., 2015). On the one hand, the speech processing system that includes the left temporoparietal cortex (TPC) and inferior frontal areas develops early in life and provides neural resources that can be recruited for acquiring reading skills later (Liebig et al., 2021; Lohvansuu et al., 2018). In line with this, individuals with dyslexia manifest impaired speech processing (Boets et al., 2007; Noordenbos & Serniclaes, 2015; Ziegler et al., 2009) and abnormal neural activity in regions including the left TPC (Landi et al., 2010; Lehongre et al., 2011; Lizarazu et al., 2020; Nora et al., 2021). Such deficits are likely to be transmitted from parent to offspring, which is partially reflected as impaired speech perception and abnormal activation of the left TPC during phonological processing in pre-literate children with a family risk of dyslexia (de Klerk et al., 2021; Mittag et al., 2021; Raschle et al., 2012).

On the other hand, learning to read also modifies phonological coding and grapho-phonological mapping via reshaping the underlying neural substrates (Dehaene et al., 2010, 2015). Dehaene et al. (2010) found that activation of the left TPC in speech perception was distinguishable between illiterate and literate adults. Higher activation of this area was also observed in 6-year-old children who could read versus those who could not (Monzalvo & Dehaene-Lambertz, 2013), and typical children versus dyslexia (Blau et al., 2009; Monzalvo et al., 2012). Moreover, activation of the visual word form area (VWFA)—a subregion located in the left occipitotemporal cortex (OTC) that responds specifically to visual words and word-like stimuli (e.g., letter strings)—in response to speech is only observed among literate adults and typically developing children (Cone et al., 2008; Dehaene et al., 2010; DeMarco et al., 2017; Vinckier et al., 2007; Wang et al., 2018, 2020). Of note, in addition to formal reading instruction, several factors affect whether activation in the left OTC can be observed, of which in-scanner task paradigms, linguistic features of stimuli (DeMarco et al., 2017), and individual differences in reading ability (Conant et al., 2020; Desroches et al., 2010; Junker et al., 2020; Liebig et al., 2021; Maier et al., 2004; Raschle et al., 2012) are major ones that should be carefully considered.

It is worth noting that while the neural signature for fluent reading is universal (Rueckl et al., 2015), linguistic characteristics, such as orthographic depth, can modulate the precise manifestations of particular components in the reading system (Richlan, 2014). Thus, testing a hypothesis developed in specific languages with other languages characterized by different linguistic features is necessary for clarifying its universality (Landerl et al., 2021). In terms of the speech-reading relationship, letters directly code speech sounds in alphabetic languages, known as grapheme-phoneme correspondence (GPC). Conquering the GPC rules and achieve the automatic level is crucial for successful reading acquisition. Even in English, a deep opaque orthography, this statement holds at the bigram level (Ziegler & Goswami, 2005). In other words, the phonological representation can be automatically accessed from the orthographic input and vice versa in skilled readers. There-

Research Highlights

- Activation of the left temporoparietal cortex in speech perception is positively associated with Chinese children's reading ability.
- Higher-skill readers recruit the left anterior fusiform gyrus more than lower-skill readers in processing pseudowords.
- Reaction time (RT)-related brain activation in the pre-cuneus varies with reading ability.
- The right hippocampus manifests a strong lexicality effect.

fore, it is not surprising to observe the neural features (e.g., activation) of the speech network, such as the left TPC, vary with reading level, and to observe orthographic-related regions, such as the left OTC, activate during speech processing and correlate with reading performance.

To date, however, most studies that investigated the brain basis of the speech-reading relationship were conducted in alphabetic languages. In Chinese, a non-alphabetic language with no GPC rules, the mapping between print and speech sound occurs at the syllable level and is entirely arbitrary. In addition, Chinese is replete with homophones. That is, one syllable corresponds to more than one visually and semantically different character. For example, the syllable /shi 4/ (the number stands for lexical tone) can be written with more than five commonly used orthographies (e.g., “是,” “世,” “市,” “氏,” “士,” “室”). Such one-to-many mapping between phonology (syllable) and orthography (character) can be solved in the compound word (e.g., /shi 4/ /chang 3/ to “市场,” meaning market), which represents approximately 80% of words in Chinese, but not in the pseudoword that consists of syllables existing in Chinese phonology but is not a real word (i.e., does not have any meanings) as a whole, where the top-down effect is lacking. In this case, answering questions about whether the phonological-related regions (e.g., left TPC) and orthographic-related regions (e.g., left OTC) are recruited during Chinese spoken word processing and whether such recruitments are associated with reading ability is necessary for examining whether the combination of phonological and orthographic processing systems is a universal mechanism underlying reading development across languages.

At the behavioral level, studies conducted in Mandarin-speaking communities have demonstrated the importance of speech processing skills (e.g., phonological awareness) in Chinese reading acquisition (Chow et al., 2005; Chung & Bidelman, 2021; Landerl et al., 2021; Li et al., 2012; Shu et al., 2008; Siok & Fletcher, 2001) and identifying children with dyslexia (Wang & Bi, 2021). At the brain level, studies have revealed impaired speech perception (e.g., categorical perception of lexical tone) in Chinese children with dyslexia (Cheung et al., 2009; Zhang et al., 2012). Specifically, dyslexic children perceived lexical tone contrasts less categorically than age-matched controls, and they did not elicit larger mismatch negativity for the cross-category tone contrast than the within-category distinction in the left hemisphere (Zhang



et al., 2012). Of note, the previous research almost focused on the temporal domain by using event-related potential (ERP), and the number of research is limited. A few studies have investigated auditory lexical processing in Chinese with functional magnetic resonance imaging (fMRI) and revealed a left-lateralized network that consists of the left superior/middle temporal areas, dorsal and ventral inferior frontal cortex, and middle frontal gyrus (Liu & Chen, 2020; Liu et al., 2009; Xiao et al., 2005). However, the question about which brain region displays such a speech-reading association remained largely open.

In terms of orthographic activation during speech processing, while behavioral and ERP studies demonstrated the orthographic effect in Chinese speech perception (Qu & Damian, 2017; Zou et al., 2012), fMRI data further revealed activation of the left OTC in both Chinese children and adults (Cao et al., 2011; Deng et al., 2013). However, knowledge about which brain regions underlie the relationship between orthographic activation during speech processing and reading ability is scarce, where more fMRI studies are needed. Zou and colleagues found that the orthographic facilitation effect on auditory word recognition is associated with reading skills in normal adults in an ERP study (Zou et al., 2012), indicating a potential role of the orthographic processing areas.

1.1 | The present study

Against this background, we directly examined the relationship between the brain response to speech and reading ability in Chinese children with fMRI. Three specific questions were asked. First, whether the brain areas for speech perception (e.g., left TPC) are associated with Chinese children's reading ability. Second, whether the orthographic processing related left OTC is activated in an auditory lexical decision task that does not require explicit phonological-orthographic conversion and whether such activation correlates with reading ability. Third, it has been shown that trial-by-trial reaction time (RT)-related brain activity can provide unique information about neural mechanisms (e.g., "neural effort") underlying given cognitive processing (Taylor et al., 2014; Yarkoni et al., 2009). Thus, we attempted to determine whether RT modulates the activation in specific brain regions; and whether such modulation is associated with individual differences in reading ability.

Specifically, an auditory lexical decision task was used. It consisted of word and pseudoword conditions, where the stimuli were disyllabic sounds, and has previously been used in Chinese adults (Xiao et al., 2005). Spoken word recognition is a continuous and dynamic process, with the meaning being derived as an acoustic input unfolds. Given that so many homophones exist in Chinese, while the phonology of a compound word is perceived sequentially, the initial syllable simultaneously activates a cohort of lexical candidates. As the acoustic signal unfolds and more information becomes available, initially activated alternatives are suppressed until only one candidate remains. However, no candidates are available after the second syllable was presented in processing pseudowords. In addition to the bottom-up acoustic input, other factors such as orthography also influence spoken word recognition as that phonology, semantics, and orthography are all inter-

connected component systems. Since only words exist in the mental lexicon, the orthography could be automatically activated for words but not pseudowords in lexical decision. Therefore, in this study we on the one hand focused on the overall activation across the word and pseudoword conditions, which mainly reflected a serial cognitive process including phonology access, orthography activation, semantic retrieval, and decision making. On the other hand, we examined the activation difference between conditions where only the stimuli in the word condition were existed in the mental lexicon and had a certain corresponding orthography.

2 | MATERIALS AND METHODS

2.1 | Participants

Twenty-five children (16 females; age range 118–143 months) were included in the brain analysis. They were selected from 31 children who attended both the MRI session administered at Beijing Normal University and the behavior session administered at local schools, where six children were excluded because of uncompleted MRI data collection ($n = 1$) or severe head motion during the functional scans ($n = 5$). These children were all right-handed, with normal hearing and normal or corrected-to-normal vision. The verbal and nonverbal intelligence quotients (IQs) measured with the Chinese Wechsler Intelligence Scale for Children (WISC-CR) (Wechsler, 1974) were larger than or equal to 85 for all children except three who did not receive the IQ test due to scheduling issues. They were part of a larger sample of 205 children (93 girls) in grades 4–6 recruited from local elementary schools. They were all native speakers of Chinese without a history of neurological/psychiatry disorders, according to the parents' reports. A battery of tests was used to evaluate reading and reading-related cognitive skills. This study was approved by the Institutional Review Board of State Key Laboratory of Cognitive Neuroscience and Learning at Beijing Normal University. All the children and their guardians were given detailed explanations about the experiment and provided signed informed consent before participating in the study.

2.2 | Reading measurements

Five tests were used to assess reading and reading-related cognitive-linguistic skills in all 205 children. (a) *Character Recognition* was used to measure character-level reading skills (Xue et al., 2013). This test contains 150 characters arranged in the teaching grade and has been widely used to identify children with dyslexia in mainland China (Cui et al., 2016; Xia et al., 2016). (b) *Word List Reading* consisted of 180 high-frequency two-character words and was used to measure oral reading fluency (Xia et al., 2018). (c) *Rapid Automatized Naming* (RAN; digit version) was used to measure efficiency while children retrieved phonological information from visual input (Liu et al., 2017). (d) *Phoneme Deletion* required the participant to delete onset, middle, or final phoneme from an orally given syllable (Li et al., 2012) and was used to assess

phonological awareness. (e) *Morphological Production* was used to measure character-level morphological awareness (Shu et al., 2006). In this task, the participant was asked to produce two new words with the target character (e.g., /mian 4/ orally presented with the original word (/mian 4/ /fen 3/; meaning “flour”). The meaning of the given character (i.e., /mian 4/) in one new word should remain the same (e.g., /mian 4/ /bao 1/; meaning “bread”) as the original one, while it in the other needs to be different (e.g., /mian 4/ /kong 3/; meaning “face”).

Given the strong correlations between reading-related measures in all 205 children (Table S1), we first reduced the dimension of behavioral matrices by using factor analysis with SPSS Statistics version 24 (SPSS Inc., Chicago, IL, USA). The results of the Kaiser-Meyer-Olkin test ($KMO = 0.745$) and Bartlett's test of sphericity (Approx. Chi-square = 437.8, $p < 0.001$) indicated that the data were suitable for factor analysis. Then, we applied factor analysis where a principal component analysis without rotation was used as the extraction method. One factor (named readFac) was identified with the criteria of eigenvalue larger than one and accounted for 59.5% of the variance. Finally, we used the regression method to calculate the readFac score for each child. To examine whether readFac generated with all 205 children's data also captured the variations in reading measures in the individuals with MRI, we calculated the correlations between the readFac and reading scores in the fMRI subsample. We also calculated the correlations between readFac and in-scanner task performance (accuracy, RT) and nuisance variables (age, sex, IQ, head motion measured by mean framewise displacement (FD)).

2.3 | Experimental design, stimuli, and procedure

In this study, we measured brain responses to speech with an auditory lexical decision task while collecting fMRI data. The word and pseudoword conditions were set up, each consisting of 30 trials. The stimuli were 30 two-character compound words with an average ($\pm$ standard deviation; SD) frequency of 15.23 (± 20.54) instances per million (Yu et al., 1998) and 30 two-character pseudowords, which were selected from a previous study (Zou et al., 2012). In brief, the pseudowords were generated by randomly combining the first and second characters from the real word stimuli. The sound files were recorded by a native Mandarin female in a soundproof room with a sampling rate of 48 kHz and 16-bit quantization. The average ($\pm SD$) durations of words and pseudowords were 708 (± 45) ms and 711 (± 46) ms, respectively. There was no significant difference between the two conditions ($t = -0.261$, $p = 0.795$). A group of adults was recruited and asked to judge the pseudowords to ensure the stimuli did not sound like real words. Another group of adults were recruited to perform the same task but without fMRI data collection. The averages ($\pm SD$) of the accuracy for words and pseudowords are 0.967 (± 0.036) and 0.967 (± 0.057). No significant difference existed between the two conditions, verifying the validity of pseudowords selection. As for RT, the average ($\pm SD$) for words and pseudowords are 1060 (± 132) ms and 1305 (± 255) ms. The participants performed significantly slower processing pseudowords than words ($t = -5.831$, $p < 0.001$).

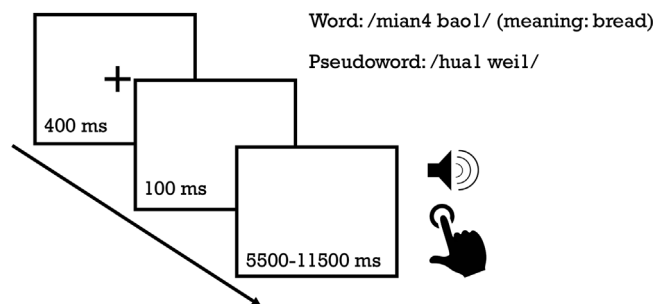


FIGURE 1 Experimental procedure (a sample trial sequence and stimuli examples) of the current fMRI study

We used a rapid event-related design in which the order of the stimuli was pseudorandomized, with no more than three of the same responses in a row. Auditory stimuli were presented via MRI-compatible earphones. Before the formal experiment, the sound volume was adjusted for each child to ensure that the stimuli could be clearly audible above the scanner noise. The presentation of stimuli and response recording were controlled with E-prime version 1.1 (Psychology Software Tools, Inc.). All the 60 stimuli (30 words and 30 pseudowords) were delivered in a single functional run. There was a 16-s silent rest period at the beginning of the run and a 6-s silent rest period at the end. In each trial (Figure 1), a fixation was presented at the center of the screen for 400 ms, followed by a 100-ms blank. Then, one stimulus was presented auditorily, followed by the second blank. The child was asked to decide whether the disyllabic sound delivered via the earphones is a real word or not by pressing corresponding buttons as accurately and quickly as possible. The response and RT (from the onset of the appearance of stimuli) were recorded. The total length from the stimulus presentation to the end of the following blank screen were 5500, 7500, 9500, or 11,500 ms. That is, there were four inter-trial intervals (6 s, 8 s, 10 s, and 12 s), which were equally and randomly assigned to each trial.

2.4 | Image collection and preprocessing

The MRI data were collected in the Imaging Center for Brain Research at Beijing Normal University using a 3-Tesla Siemens MAGNETOM Trio Tim scanner with a 12-channel head coil. Before scanning, each child was given a detailed introduction about the experiment and got familiar with the environment and scanner noise. In the scanner, foam pads were used to help reduce head motion. One functional run ($T2^*$ sensitive gradient echo planar imaging) and one $T1$ -weighted image (magnetization-prepared rapid acquisition with gradient echo) were collected for each child. The parameters of the functional sequence were as follows: repetition time (TR) = 2000 ms, echo time (TE) = 30 ms, flip angle (FA) = 90 degrees, slice thickness = 3.0 mm, voxel size = $3.0 \times 3.0 \times 3.0 \text{ mm}^3$, interscan gap = 0.485 mm, number of slices = 42, number of volumes = 278, and time of acquisition = 9 min and 16 s. The parameters of the $T1$ image were as follows: TR = 2530 ms, TE = 3.45 ms, FA = 7 degrees, slice thickness = 1.0 mm,



voxel size = $1.0 \times 1.0 \times 1.0 \text{ mm}^3$, interscan gap = 0 mm, number of axial slices = 176, time of acquisition = 8 min and 6 s. After images were collected, a radiologist who was blinded to the details of the study reviewed the quality and ascertained any pathological deviations immediately.

SPM12 (version 7771; Wellcome Department of Cognitive Neurology, London, UK, <http://www.fil.ion.ucl.ac.uk/spm/>) was used to analyze the neuroimaging data. The first four volumes of the functional run were discarded to allow the MRI signal to reach equilibrium. Then, motion correction and slice timing correction were administered. Brain volumes with a frame-to-frame head motion larger than 2 mm or an intensity larger than 9 SD from the global mean were labeled as outliers and used as covariates in the subsequent hemodynamic response function (HRF) estimation (https://www.nitrc.org/projects/artifact_detect). In addition, the average FD was calculated and used as a covariate in the group-level analysis. Children who had an average FD larger than 0.5 mm were excluded from the subsequent analysis. The resultant images were normalized to the standard template in Montreal Neurological Institute (MNI) space via T1 images. Finally, the images were smoothed with an isotropic 8-mm full-width at half-maximum Gaussian kernel.

At the individual level, 16 regressors and timing points for outlier brain volumes were included in HRF estimation. Specifically, brain activations under the word and pseudoword conditions were modeled separately. We also separately modeled the corrected and uncorrected trials, as well as those without response. Trial-by-trial RTs were included as parametric regressors to identify RT-related activations for the word and pseudoword conditions. Uncorrected responses and responses out of range (less than 100 ms or larger than 6000 ms) were excluded from the analysis by setting separate regressors. Motion parameters and timing files of the outlier volumes (mean = 2; range: 0–12) were also included in the model as covariates.

2.5 | Statistics

2.5.1 | Analytic approach

The main aim of this study was to examine brain responses to speech and their correlations with reading performance in Chinese children. Since the experiment consisted of word and pseudoword conditions, we focused on both activation across the entire task (i.e., combining the word and pseudoword conditions) and activation differences between conditions (i.e., the word condition against pseudoword condition). Four contrasts were of interest: (i) Activity_{word+pseudoword} was defined as the average brain activation across the word and pseudoword conditions; (ii) Activity_{word-pseudoword} was defined as the difference in brain response between the word condition and pseudoword condition, which was also named as the lexicality effect in previous literature; (iii) RT_{word+pseudoword} was defined as the brain response modulated by response latency across the word and pseudoword conditions, and it revealed areas associated with neural effort spent across the entire task; (iv) RT_{word-pseudoword} was defined as the difference in response

latency-modulated brain responses between the word condition and pseudoword condition, and it revealed areas distinctively associated with neural efforts spent between the word and pseudoword conditions. For different contrasts, we identified brain areas showing activation across conditions and differences between conditions by using one-sample *t*-tests and paired sample *t*-tests and detected areas with significant brain-behavior correlations by using regression analyses.

We conducted both voxel-wise whole-brain analyses and region-of-interest (ROI) analyses, which complement each other. The voxel-wise analysis was performed using SPM12, while the ROI analysis was conducted using SPSS.

2.5.2 | Voxel-wise whole-brain analysis

In the analysis at the voxel level, we examined brain activation and brain-behavior correlations for each contrast. All the analyses were performed within a gray matter mask created using all the children's gray matter segmentation images with a cutoff value of 0.2. A threshold of cluster-forming *p*-voxel < 0.001, family-wise error (FWE)-corrected *p*-cluster < 0.05 was used to identify significant clusters for the contrasts (i) Activity_{word+pseudoword} and (ii) Activity_{word-pseudoword}, and to examine brain-behavior correlations for all contrasts. Because the RT modulation explained variations in the signal that were not accounted for by word/pseudoword conditions, we used a liberal voxel-level threshold *p* < 0.01 in detecting RT-related areas (i.e., contrasts (iii) RT_{word+pseudoword} and (iv) RT_{word-pseudoword}). Nevertheless, the cluster-level *p*-value was still corrected with the FWE method implemented in SPM, and only regions with a corrected *p*-cluster < 0.05 were reported. For each significant cluster in the brain-behavioral correlation analysis, a follow-up analysis was conducted on the average estimated value of the cluster to examine whether the effect was driven by any nuisance variables (i.e., age, sex, and head motion).

Significant results were presented on a FreeSurfer template surface with BrainNet Viewer (<https://www.nitrc.org/projects/bnv/>). In addition, information about the significant clusters was reported with DPABI (<http://rfmri.org/dpabi>).

2.5.3 | ROI definition and analysis

In the ROI analysis, we examined brain areas that are involved in either speech perception (Davis & Gaskell, 2009; Gow, 2012) or reading processing (Taylor et al., 2013), including the left superior temporal gyrus, supramarginal gyrus, angular gyrus, left inferior frontal cortex (pars opercularis and pars triangularis), left OTC, and bilateral hippocampus. All the ROIs except the left OTC were defined using the automated anatomical atlas (AAL) (Rolls et al., 2020) with the tools implemented in DPABI, and were labeled as LHES, LSTG, LSMG, LANG, LIFGoperc, LIFGtriang, LHIP, and RHIP. Because the neurofunctional property in the left OTC largely depends on the location (e.g., anterior vs. posterior subregions), we defined 6 ROIs in the left fusiform gyrus (LFG) according to the method of Vinckier et al. (2007). The ROIs were

defined as spheres center at loci from posterior to anterior: LFG-1 (MNI coordinates $x y z = -18 -96 -10$), LFG-2 ($-36 -80 -12$), LFG-3 ($-46 -64 -14$), LFG-4 ($-48 -56 -16$), LFG-5 ($-50 -48 -16$), and LFG-6 ($-50 -40 -18$), with a radius of 4 mm. LFG-4 corresponds to the classic VWFA. Since there were 14 ROIs in each analysis, multiple comparisons were corrected using the Bonferroni method (corrected $p < 0.05/14 = 0.00358$). In the results section, uncorrected p values were reported and compared with the corrected threshold.

2.5.4 | Supplemental analysis

Given that two children in the primary data analysis performed below the chance level in the auditory lexical decision task, we conducted two supplemental analyses to investigate the influence of the low task performance. In the first analysis, we would like to be strict and attempted to control children's ability of making auditory lexical decisions. Thus, we excluded children with an ACC lower than 0.5 and repeated the entire data analysis steps. The same FWE-corrected threshold of p -cluster < 0.05 and the Bonferroni method were used for controlling multiple comparison errors in the voxel-wise whole-brain analysis and predefined ROI analysis. Of note, if a cluster identified in the primary analysis was not observed at the whole-brain level, ROI analysis would be run in this cluster. Alternatively, the neuro-cognitive processing accompanying correct and incorrect responses could be the same. Therefore, in the second analysis, we included those two children with poor in-scanner performance and pooled together all the trials with a response within the acceptable window (i.e., no matter whether the children's answer was correct or not). Similarly, the entire analysis steps were repeated.

3 | RESULTS

3.1 | Behavioral results

Table 1 presents the mean and SD values of the reading measures, accuracies (ACCs) and response latencies (RTs) for correct trials of the in-scanner task. As expected, children showed lower ACCs and longer RTs for both conditions than adults. But same as adults, children performed faster in the word condition than in the pseudoword condition ($t = -4.31$, $p < 0.001$) and presented no significant difference between the word and pseudoword conditions on ACC ($t = -0.275$, $p = 0.785$). Reading measures were significantly correlated with each other (p 's < 0.05) except that between phoneme deletion and RAN ($r = 0.26$, $p = 0.206$). Importantly, readFac (composite score obtained from factor analysis) was significantly correlated with all the reading measures in the fMRI sample (character recognition: $r = 0.866$, $p < 0.001$; oral reading fluency: $r = 0.952$, $p < 0.001$, RAN: $r = 0.760$, $p < 0.001$; phoneme deletion: $r = 0.647$, $p < 0.001$; morphological production: $r = 0.789$, $p < 0.001$). Therefore, we used readFac as a proxy of overall reading ability in the subsequent analyses to explore brain-behavior correlations. Consistent with previous literature, read-

TABLE 1 Demographics and behavioral measures of children in fMRI analysis

Variables	Mean	SD
Demographics		
N	25	—
Age (month)	132.0	7.0
Sex (female %)	64	—
General Abilities ^a		
Verbal IQ (standard score)	112.6	14.7
Performance IQ (standard score)	119.6	11.5
Full-scale IQ (standard score)	117.2	12.4
Reading skills		
Character recognition (item)	113.6	20.6
Word list reading (word/minute)	93.7	25.3
Rapid naming (item/second)	13.6	2.9
Phoneme deletion (item)	11.0	3.2
Morphological production (item)	21.8	4.2
In-scanner task		
Accuracy _{Overall}	0.63	0.09
Accuracy _{Word}	0.63	0.16
Accuracy _{Pseudoword}	0.64	0.16
Reaction Time _{Overall} (ms)	1685	343
Reaction Time _{Word} (ms)	1558	300
Reaction Time _{Pseudoword} (ms)	1883	451

^a $n = 22$.

Fac was associated with verbal IQ ($r = 0.661$, $p < 0.001$) but not non-verbal IQ ($r = 0.295$, $p = 0.182$). To examine the effect of nuisance variables, we performed a supplemental analysis and found no significant correlations between the readFac and in-scanner performance for either condition (ACC_{word}: $r = 0.302$, $p = 0.142$; ACC_{pseudoword}: $r = -0.127$, $p = 0.546$; RT_{word}: $r = -0.137$, $p = 0.513$; RT_{pseudoword}: $r = -0.029$, $p = 0.891$) or with nuisance variables (age: $r = 0.023$, $p = 0.912$; sex: $r = -0.033$, $p = 0.876$; mean FD: $r = -0.283$, $p = 0.171$). Finally, a positive correlation between ACC_{word} and phoneme deletion score was observed ($r = 0.51$, $p < 0.01$).

3.2 | Brain results

3.2.1 | Voxel-wise whole-brain analysis

Across the entire task, widespread areas over the temporal, frontal, and occipital cortices, including the left OTC, were activated (FWE-corrected p -cluster < 0.05 ; Figure 2a, Table 2). The bilateral insula and premotor cortex were significantly modulated by trial-by-trial RT (Figure 2b, Table 2). No region showed a significant difference between the word condition and the pseudoword condition in terms of task-induced activation or RT-related activation at the whole-brain corrected threshold.

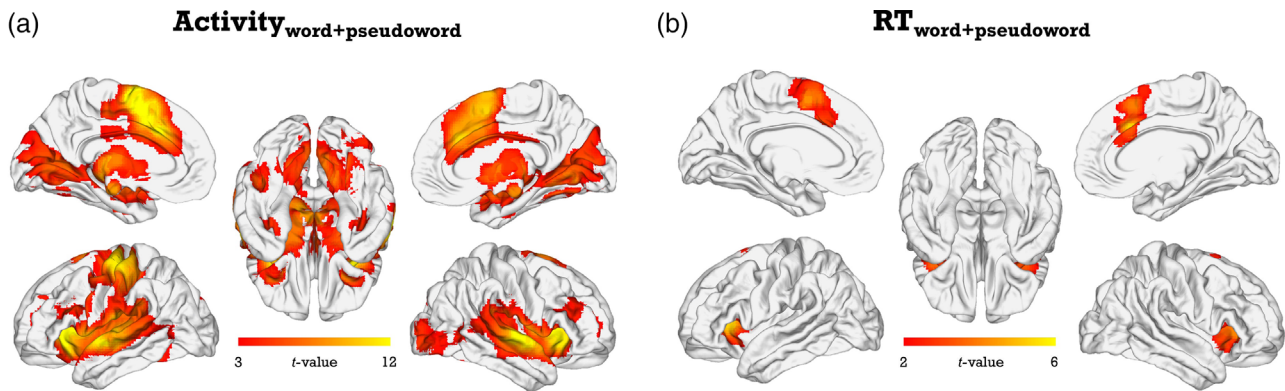


FIGURE 2 Whole-brain analyses of activation and reaction time (RT) modulation. Significant clusters were identified based on a threshold of FWE-corrected p -cluster < 0.05 . (a) Brain areas were significantly activated across the word and pseudoword conditions. The contrast is labeled as Activity_{word+pseudoword}. (b) Brain areas were significantly related to trial-by-trial RT. The contrast is marked as RT_{word+pseudoword}. No clusters showed significant differences between conditions in brain activation or RT modulation at the whole-brain threshold

In the voxel-wise regression analysis, overall activation of a cluster located in the left TPC was positively correlated with the readFac value (FWE-corrected p -cluster < 0.05 ; Figure 3a left panel, Table 2). The post hoc analysis revealed significant brain-reading associations in both the word and pseudoword conditions (Figure 3a middle and right panels). The correlations remained significant when the nuisance variables (age, sex, mean FD) were statistically controlled (word: partial $r = 0.754$, $p < 0.001$; pseudoword: partial $r = 0.686$, $p < 0.001$). In terms of RT_{word-pseudoword}, a cluster in the precuneus showed a negative correlation with readFac (Figure 3b left panel). The post hoc analysis revealed that this effect was mainly driven by a negative correlation between reading and RT-related brain activation in the word condition (Figure 3b middle and right panels). The precuneus's activity in response to real words was more associated with trial-by-trial RT variance in children with lower reading skills. No clusters showed a significant correlation between reading and Activity_{word-pseudoword} or RT_{word+pseudoword} at the FWE-corrected threshold.

3.2.2 | ROI analysis

The predefined ROIs are displayed on standard brain templates (Figure 4a and b). Here, we summarized the findings of FG ROIs and the other ROIs (hereafter named language/reading ROIs) separately for the aim of clarity. First, almost all the language/reading ROIs were significantly activated during the entire auditory lexical decision task (p 's $< 0.05/14 = 0.00358$; Bonferroni correction) except LSMG and LANG (Figure 4a right panel). Second, in comparing the word and pseudoword conditions, only one ROI survived the Bonferroni correction: activation in RHIP in response to real words was significantly higher than pseudowords. Third, in the fusiform gyrus, LFG-3 and LFG-4 showed significant activation for the word condition (p 's $< 0.05/14 = 0.00358$; Bonferroni correction; Figure 4b middle panel), no activations were found in the pseudoword condition. Finally, no significant differences were found between the word and pseudoword conditions in FG ROIs.

The analyses of brain-behavior relationships showed a negative correlation between the lexicality effect in LFG-6 and readFac value ($p < 0.05/14 = 0.00358$; Bonferroni correction; Figure 4b right panel). The post hoc analysis revealed a positive correlation between the brain activation to pseudowords and reading scores ($r = 0.423$, $p = 0.049$), while no significant correlation was observed in the word condition ($r = -0.215$, $p = 0.338$).

Finally, none of the ROIs showed significant RT-related activity, differences in RT-related activations between conditions, or significant correlations with reading scores.

3.3 | Supplemental analysis

In the first analysis, we excluded two children due to their chance level in-scanner task performance. The main findings of the primary analysis were almost replicated and are summarized as follows; (1) The widespread activations across the frontotemporal cortex were observed (Figure S1A), and the left OTC was also activated. (2) In terms of RT-related activation, although we did not find the salience network survived the whole threshold (Figure S1B), ROI analysis based on the primary analysis reveal a positive association between the activation of the bilateral insula and premotor area with trial-by-trial RT (Table S2). (3) No cluster displayed a significant difference between conditions on activation strength or RT modulation. (4) Significant correlations between brain measures and reading ability were observed at the ROI level (Table S3). In the predefined language/reading ROI analysis, the main finding was also replicated. (5) We observed activation of almost all language/reading ROIs and middle portion of the fusiform gyrus (i.e., LFG-3, LFG-4) in speech processing, especially in the word condition (Table S4). (6) The right hippocampus displayed a strong lexicality effect, which survived the Bonferroni correlation (Table S4). (7) The lexicality effect in the FG-6 was correlated with readFac ($r = -0.503$, $p = 0.015$), which was mainly driven by the pseudoword condition ($r = 0.436$, $p = 0.038$) but not the word condition ($r = -0.017$, $p = 0.938$). However, the correlations were weaker with

**TABLE 2** Significant clusters identified in the whole-brain analyses

Contrast	Brain area (AAL atlas; the number of voxels > 10)	Size	Peak T	MNI coordinates		
(i) Activation _{word+pseudoword}						
Activation	Temporal_Sup_R, Precentral_L, Postcentral_L, Temporal_Sup_L, Insula_L, Supp_Motor_Area_L, Cerebelum_6_R, Insula_R, Cerebelum_6_L, Temporal_Mid_L, Supp_Motor_Area_R, Calcarine_L, Cerebelum_8_R, Lingual_L, Putamen_L, Temporal_Mid_R, Calcarine_R, Putamen_R, Rolandic_Oper_L, Cingulum_Mid_R, Rolandic_Oper_R, Cingulum_Mid_L, Caudate_R, Thalamus_L, Frontal_Inf_Tri_L, Cerebelum_4_5_R, Cerebelum_8_L, Cingulum_Ant_L, Thalamus_R, Cerebelum_Crus1_L, Cerebelum_Crus1_R, Frontal_Inf_Orb_L, Caudate_L, Vermis_4_5, Temporal_Pole_Sup_L, Cingulum_Ant_R, Temporal_Pole_Sup_R, Cerebelum_4_5_L, Frontal_Inf_Oper_L, SupraMarginal_L, Hippocampus_L, Frontal_Inf_Orb_R, Vermis_6, Cuneus_L, Cerebelum_Crus2_R, Hippocampus_R, Frontal_Sup_Medial_L, Frontal_Inf_Tri_R, Occipital_Inf_R, Frontal_Sup_L, Cerebelum_Crus2_L, Heschl_R, Frontal_Inf_Oper_R, SupraMarginal_R, Heschl_L, Parietal_Inf_L, Frontal_Sup_Medial_R, Fusiform_R, Cuneus_R, Vermis_8, Vermis_7, Amygdala_L, Occipital_Mid_R, Amygdala_R, Fusiform_L, Temporal_Inf_L, Cerebelum_7b_R, Cerebelum_7b_L, Frontal_Mid_R, Pallidum_R, Cerebelum_9_R, Pallidum_L, ParaHippocampal_R, Postcentral_R, Frontal_Mid_L, Temporal_Inf_R, Precuneus_L, Olfactory_L, ParaHippocampal_L, Frontal_Sup_R, Precuneus_R, Cerebelum_3_R, Paracentral_Lobule_L, Vermis_9, Olfactory_R, Vermis_3, Occipital_Mid_L, Temporal_Pole_Mid_R, Occipital_Inf_L, Frontal_Sup_Orb_L	49827	14.20	−10	−4	+60
	Frontal_Mid_R, Frontal_Sup_R	262	4.65	+32	+50	+4
	Frontal_Mid_L	243	6.14	−26	+42	+12
Correlation	SupraMarginal_L, Temporal_Sup_L, Temporal_Mid_L	250	5.34	−54	−28	+30
(ii) Activation _{word-pseudoword}						
Activation	n.s.					
Correlation	n.s.					
(iii) RT _{word+pseudoword}						
Activation	Insula_L, Putamen_L, Frontal_Inf_Tri_L, Frontal_Inf_Oper_L, Pallidum_L, Frontal_Inf_Orb_L	935	5.26	−30	+24	+4
	Insula_R, Putamen_R, Frontal_Inf_Tri_R, Frontal_Inf_Oper_R, Caudate_R, Pallidum_R	922	4.09	+30	+26	+2
	Supp_Motor_Area_L, Cingulum_Mid_R, Supp_Motor_Area_R, Cingulum_Ant_R, Cingulum_Mid_L, Frontal_Sup_Medial_L, Frontal_Sup_Medial_R	1182	5.44	+10	+20	+30
Correlation	n.s.					
(iv) RT _{word-pseudoword}						
Activation	n.s.					
Correlation	Precuneus_L, Precuneus_R, Paracentral_Lobule_R, Postcentral_L, Parietal_Sup_R, Precentral_L	1395	5.57	+12	−34	+64

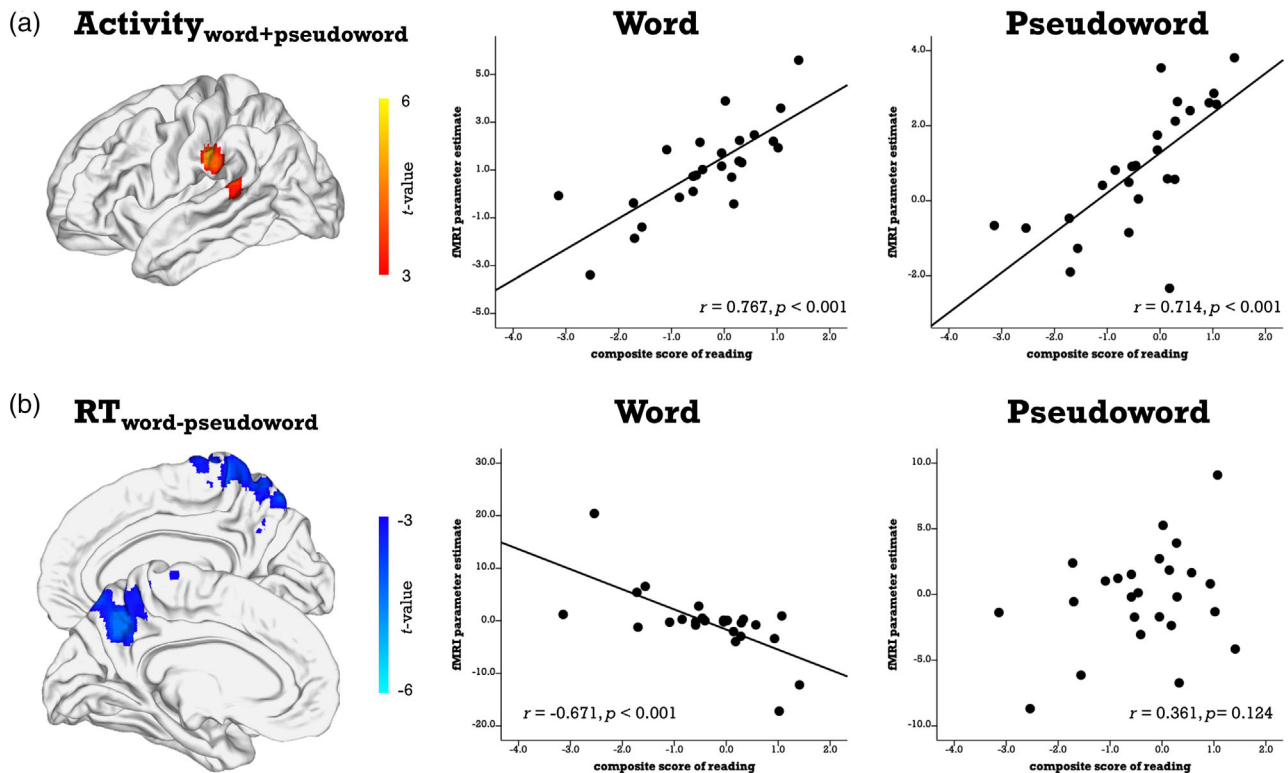


FIGURE 3 Whole-brain analyses of brain-behavioral correlations. (a) Left panel: cluster showing a significant correlation between the activation across conditions and composite scores of reading (FWE corrected p -cluster < 0.05); middle panel: scatter plot of the correlation between activation of the left TPC (red) in the word condition with reading scores; right panel: scatter plot of the correlation between activation of the left TPC cluster in the pseudoword condition with reading scores. (b) Left panel: cluster showing a significant correlation between the difference on trial-by-trial reaction time (RT) modulated activation between word and pseudoword conditions with reading ability (FWE-corrected p -cluster < 0.05); middle panel: scatter plot of the correlation between the modulated activation of the precuneus (blue) in the word condition with reading scores; right panel: scatter plot of the correlation between the modulated activation of the precuneus in the pseudoword condition with reading scores. No clusters showed a significant correlation between reading scores and activation difference between the word and pseudoword conditions or RT-modulated activation across conditions at the whole-brain threshold

the smaller sample and could not survive the Bonferroni correction. (8) No language/reading ROI showed RT-related activation or correlation between RT-related activation and readFac.

The second analysis included all the children with qualified imaging data and pooled together all the children's responses with an appropriate RT. The results are like the first supplementary analysis and the original findings (Figures S1C and D, S2; Table S3 and S4), except the activations of areas in the salience network were not correlated with RT (Table S2), indicating a potential difference in "neural effort" between correct and incorrect responses.

4 | DISCUSSION

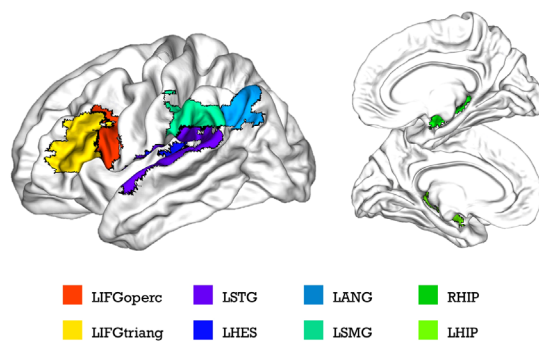
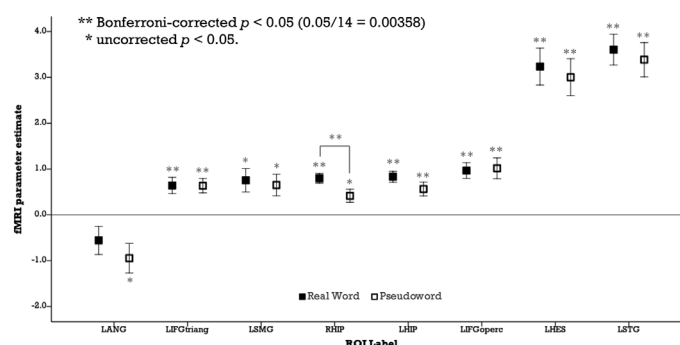
In the present study, we examined the brain activations accompanying auditory lexical decisions and whether such neural responses were correlated with reading ability in Chinese children. The results collectively demonstrate a concrete speech-reading association. First, while we found widespread activations in the speech network such as the left TPC, an area in the left OTC near the classic VWFA was significantly activated, especially in the word condition. Furthermore, acti-

vations of the left TPC in both the word and pseudoword conditions were strongly correlated with children's reading ability. Second, the right hippocampus displayed a striking lexicality effect, that is, stronger activation in the word condition than the pseudoword condition. Third, we found a negative correlation between lexicality and reading performance in the left anterior OTC. The subsequent analysis revealed children with higher reading scores showed stronger activation in this area in processing pseudowords. Finally, activation in the bilateral insula and premotor cortex was associated with trial-by-trial response latency. Such RT-related neural response in the precuneus and surrounding areas was negatively correlated with reading ability.

4.1 | The speech processing related left TPC and reading

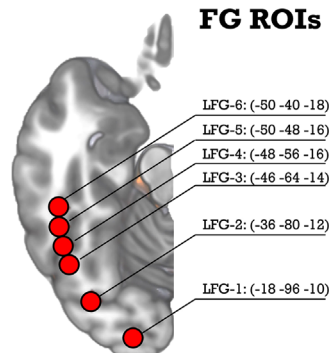
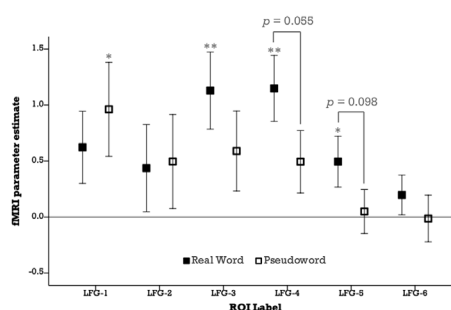
The left TPC has been proposed to be important in both speech perception and reading because of its central role in phonological processing across languages (Cattinelli et al., 2013; Chang et al., 2010; Feng et al., 2018; Hodgson et al., 2021; Liu et al., 2009; Si et al., 2017; Yan et al., 2021). Lesions in the left TPC caused overall and non-selective deficits

(a) Language/Reading ROIs

Mean $\pm$ S.E.M.

(b)

FG ROIs

Mean $\pm$ S.E.M.

LFG-6 (-50 -40 -18)

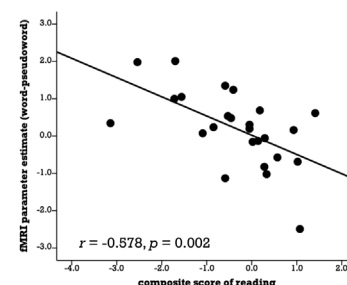


FIGURE 4 Analyses on the predefined regions of interest (ROIs). (a) Left panel: 8 ROIs were defined with the AAL atlas according to previous findings on brain areas involved in speech perception and reading; right panel: Errorbar plot showing the mean ($\pm$ S.E.M.) activation of the language/reading ROIs in the word and pseudoword conditions, as well as differences between conditions. (B) Left panel: 6 ROIs were created from posterior-to-anterior along the left fusiform gyrus with the same approach as in Vinckier et al. (2007); middle panel: Errorbar plot showing the mean ($\pm$ S.E.M.) activation of LFG ROIs in the word and pseudoword conditions as well as differences between conditions; right panel: scatter plot of correlation between different activation between word and pseudoword conditions in LFG-6 with reading. S.E.M. = standard error of the mean, ** Bonferroni-corrected $p < 0.05$ ($0.05/14 = 0.00358$), * uncorrected $p < 0.05$

in reading aloud (Dickens et al., 2021) and impeded the development of the typical neural substrate of orthography processing (Li et al., 2021). In normal populations, longitudinal studies further revealed that neural responses to speech sound at the earlier stages (including infant, preschooler, etc.) predicted reading performance later at school (Liebig et al., 2021; Lohvansuu et al., 2018; Nora et al., 2021). In addition, anomalies in the left TPC are the most consistent findings in dyslexia research (Blau et al., 2009; Gabrieli, 2009; Hoeft et al., 2006; Monzalvo et al., 2012; Paz-Alonso et al., 2018; Richlan et al., 2011, 2013). Recently, Lizarazu and colleagues found that the neural response to speech edges in the left temporal regions was weaker in dyslexic individuals and might lead to phonological deficits (Lizarazu et al., 2020).

Unlike alphabetic languages, the Chinese writing system has no direct or systematic correspondence between phonological and orthographic units. Moreover, previous findings of dyslexia among Chinese-speaking individuals are somewhat different from those among alphabetic-speaking individuals and less consistent. For example, Siok and colleagues found that the left middle frontal gyrus showed substantial between-group differences (Siok et al., 2004). Therefore, it is necessary to clarify the role of phonological processing in Chinese

reading acquisition and the corresponding neuro-cognitive mechanisms. The increasing number of studies support the importance of phonological processing in Chinese reading development (Chow et al., 2005; Chung & Bidelman, 2021; Landerl et al., 2021; Liet al., 2012; Shu et al., 2008; Siok & Fletcher, 2001; Wang & Bi, 2021). For example, researchers found that children with dyslexia manifested deficits in speech perception (e.g., lexical tones were less categorically perceived) (Cheung et al., 2009) and showed abnormal neural responses (Zhang et al., 2012). Neural responses to speech in Chinese preschoolers could also predict their reading outcomes at school (Hong et al., 2018). Furthermore, phonological processing has been revealed to significantly classify individuals with and without dyslexia (Wang & Bi, 2021). Thus, the before-mentioned inconsistency can be most likely explained by the multiple deficit model (Pennington, 2006, 2012), which proposes that a specific single deficit cannot explain all cases of dyslexia, and explained by the developmental perspective, which indicates that the effect of a particular cognitive skill on reading acquisition is a function of the developmental stage (Bai et al., 2020; Ferrer et al., 2007). Since multiple factors and cognitive-linguistic skills work together in reading development, deficits of any component may lead to reading problems. Combining the previous literature and current



findings, we argue that phonological processing is one of the most critical components in Chinese reading development and is involved from the earliest stage.

Recently, a cross-cultural neuroimaging study provided evidence for universal neural signatures of fluent reading (Rueckl et al., 2015). The authors convincingly confirmed speech-print convergence in temporoparietal and frontal areas as the shared neural feature of fluent reading across four contrasting languages—Spanish, English, Hebrew, and Chinese—with multiple analytic approaches. The findings directly support the idea that the neural circuit for spoken language is recruited in reading acquisition, regardless of particular forms of speech sound and written script. However, while the association between neural measures and reading ability has been revealed in alphabetic languages (Preston et al., 2016), evidence in Chinese children is lacking. In the present study, we found Chinese children activated widespread areas across the bilateral frontotemporal cortex during an auditory lexical decision task, in line with the previous literature (Liu & Chen, 2020; Liu et al., 2009; Xiao et al., 2005). Strikingly and more importantly, brain activation of the left TPC during speech perception was strongly and positively associated with reading ability, regardless of lexical status of the stimuli in Chinese children aged 9–12 years. Since the words and pseudowords used in this study contained equivalent phonological information at the syllable level but differed in whole word meaning, positive correlations in both conditions indicate that phonological processing is most likely to be the corresponding cognitive mechanism. This finding is in line with the reading universals hypothesis. It provides direct evidence that the neural basis underlying speech, especially phonological processing, is responsible for the large variances of individual differences in Chinese reading. This pattern is also in line with a study on the development of this region, where 5th-grade pupils showed increased activation during an auditory rhyming task in Chinese compared with those in 3rd-grade pupils (Cao et al., 2011). However, the robust correlation observed after controlling for age and sex in the current study indicates that the left TPC is more likely related to reading ability itself rather than general development. It should be kept in mind that in addition to the influence of speech on reading, reading acquisition also reshapes the neural systems underlying speech perception (Dehaene et al., 2010, 2015), which can be another factor that drives the correlation between brain activation and reading abilities found in the left TPC.

4.2 | The orthographic related left OTC in speech processing

Besides phonological processing, orthographic information is activated and affects an individual's performance in speech processing tasks (Chereau et al., 2007; Cone et al., 2008; Dehaene et al., 2010; Ziegler & Muneaux, 2007). ERPs have been used to investigate when orthographic inconsistency effects occur during spoken word processing (Perre & Ziegler, 2008). fMRI studies also revealed orthographic intrusion effects during spoken word processing, marked by the involvement of the left OTC (Booth et al., 2002, 2004; Cone et al., 2008; Vinck-

ier et al., 2007), of which the activation is associated with orthographic processing in reading (Paz-Alonso et al., 2018). Of importance, the activation of VWFA can be observed in adults with reading experience but not in illiterate adults, which may reflect access to orthographic representation during speech processing (Dehaene et al., 2015). It has also been revealed that children aged 9 but not 6 activate VWFA during native spoken language processing (Liebig et al., 2021; Monzalvo & Dehaene-Lambertz, 2013). Consistent with these findings, the activation of the left OTC during speech has been repeatedly demonstrated in both school-age children and adults under explicit requirements such as spelling and rhyming (Cone et al., 2008; DeMarco et al., 2017; Wang et al., 2018, 2020), and reduced activation was found in pre-literate children and those with dyslexia (Conant et al., 2020; Desroches et al., 2010; Liebig et al., 2021; Raschle et al., 2012).

Even in native speakers of Chinese—a non-alphabetic language that lacks a direct link between speech and print, the activation of orthographic information has been observed at the behavioral and neurophysiological levels (Chen et al., 2016; Qu & Damian, 2017; Zou et al., 2012). For example, Zou and colleagues found that sharing orthographic information can facilitate spoken language processing in Chinese, thus reducing the amplitude of the N400 component (Zou et al., 2012). Moreover, a positive correlation between such an orthographic facilitation effect and individuals' reading performance was observed, especially in tasks requiring mapping from visual inputs to phonology/semantics. In some fMRI studies, researchers also observed activation of the left OTC in speech processing (Cao et al., 2011; Deng et al., 2012), similar to that in alphabetic language. However, few studies have directly examined the brain-reading correlation in this region in Chinese.

In the present study, the whole-brain analysis revealed the activation of the left OTC during auditory lexical processing, in line with the previous literature (Cao et al., 2011; Deng et al., 2012). Moreover, the predefined ROI analyses further identified an area near the classic VWFA (LFG-3, LFG-4; i.e., the middle part of the left fusiform gyrus) robustly activated in the word condition. However, while no activations in the pseudoword condition were found, the differences between the word and pseudoword conditions were non-significant. Nevertheless, this finding suggests that this subregion of the left OTC may contain orthographic representations for all compound words, accessed through a top-down automatic mechanism (Zou et al., 2012). Unexpectedly, we did not find a significant association between activation and reading ability in these regions. But such lack of association is in line with previous developmental research. No differences in VWFA activation were observed between children in 3rd- and 5th-grades during an auditory task without the explicit requirement of orthographic processing (Cao et al., 2011). Interestingly, we observed a negative correlation between the lexicality effect in an anterior part of the left OTC (MNI coordinates $x\ y\ z = -50\ -40\ -18$) and reading scores. This correlation was mainly driven by the positive correlation between reading and brain activation in the pseudoword condition. That is, individuals with higher reading skills showed higher activation in this region in the pseudoword condition. Compared with the middle and posterior portions, the anterior part of the left OTC showed increasing



representations for both abstract orthography and phonology (Zhao et al., 2017) and higher-level semantic processing (Hodgson et al., 2021). Thus, one possible explanation is that children with higher reading abilities searched more orthographic/phonological/semantic representations associated with the auditory syllables presented in pseudowords during lexical judgment than poorer readers, which may be due to a better representation or a more efficient connection between relevant regions.

4.3 | RT-related brain activity and its relationship with reading

In addition to brain activation across conditions and difference between conditions, we investigated the neural responses associated with trial-by-trial task RT and their association with children's reading ability. In line with previous studies on general task settings (Yarkoni et al., 2009), we observed cingulo-opercular brain areas (bilateral anterior insula, premotor cortex), of which the activation was correlated with latencies across conditions regardless of reading performance. This finding suggests that the salience network plays a role in attention maintenance and target stimuli detection or refers to the so-called "neural effort" proposed by Taylor et al. (2014). Interestingly, such "neural effort" in the precuneus—a region outside the salience network—varied with reading ability. For children with lower reading abilities, activation of the precuneus was modulated more by trial-by-trial RT in the word condition. The precuneus is part of the default mode network and involved in semantic processing during auditory lexical decisions (Junker et al., 2020). The current finding thus indicates that lower-skill readers may rely more on semantic-related processing in judging real words, thus reflecting a different strategy than higher-skill readers.

4.4 | Limitations and future directions

In this study, we observed the activation of the left OTC, especially the middle part, in processing auditory real words. However, we could not elucidate whether it refers to orthographic access or other linguistic processes with the current dataset. Second, while the previous studies repeatedly revealed impaired speech perception in dyslexia, we did not find significant associations between performance in auditory word recognition and reading ability. This negative result could be caused by multiple factors, including small sample size and noisy listening conditions (Ziegler et al., 2009). However, we did observe a positive correlation between accuracy in the word condition and phonological awareness. Third, the sample size of this study is small, thus weaken our capacity to detect effects-of-interest in specific areas in a whole-brain voxel-wise manner, such as lexicality in the classic language/reading network. However, we observed such an effect in the hippocampus and its association with reading abilities in the anterior fusiform gyrus. Future studies with larger samples should compare children with and

without dyslexia and include tasks with multiple noise levels to address these issues.

Besides, more investigations can be conducted based on the current findings. For instance, examining whether and how neural responses to speech in Chinese preschool children predict their future reading outcomes at school will deepen our knowledge about reading acquisition and provide insights into the early diagnosis of dyslexia and effective intervention design. Another finding worthy of further investigation is the striking lexicality effect in the right hippocampus. While supporting episodic memory is one of the main functions of the hippocampus, the evidence about its role in reading is lacking. One recent study suggests this area may be involved in semantic-related processing (Paz-Alonso et al., 2018). In addition, the hippocampus has also been found to be involved in concept learning by establishing abstract category representation (Bowman & Zeithamova, 2018; Bowman et al., 2020). Moreover, it plays a role in learning visual-verbal associations, such as letter-sound and object-name pairs (Taylor et al., 2014). Thus, the higher activation in response to words compared with pseudowords may indicate the semantics/concept/events associated with a given word recurs in the brain. Such possibilities should be elucidated systematically in the future. Finally, a tight correlation between the left TPC activation during spoken sentence perception and white matter properties of the arcuate fasciculus posterior segment (Thiebaut de Schotten et al., 2014). Thus, it is reasonable to expect a function-structure association, which is worth demonstrating.

5 | CONCLUSIONS

This study demonstrates that the brain response to speech is associated with reading abilities in Chinese children, especially in the left TPC, which is mainly involved in phonological processing, and the anterior occipitotemporal area, which is likely to be an interface that integrates orthographic, phonological, and semantic information in language processing. This finding supports the notion that neural circuits for speech perception are recruited and essential in reading acquisition.

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CONFLICT OF INTEREST

The authors declare no competing financial interests.



ETHICS STATEMENT

This study was approved by the Institutional Review Board of State Key Laboratory of Cognitive Neuroscience and Learning at Beijing Normal University.

DATA AVAILABILITY STATEMENT

The data supporting the present study's findings are available from the corresponding author upon reasonable request.

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